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Teaching Design Ethics Through Tabletop Game Prototyping

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Abstract: *Design Ethics is a challenging yet vital topic to address and teach, especially in an age where designers may think they are not ethically responsible for their creations. This paper describes the design of an ethics course where instructors used Research Through Design as a vehicle to teach design ethics to a class of master's students. We describe how we introduced tabletop game prototyping using the Design Games Framework, a lecture-workshop-supervision format supported by a design diary, and student-led sessions to create a pedagogical structure for students to unpack and explore ethical issues. We discuss the design diaries and present reflections on developing reflexivity through game making, framing design as exploration rather than solution-making, and moving beyond dramatic elements when game-making. The course design and reflections provide guidance for design educators in integrating game-making activities in design ethics courses.*

Keywords*: design ethics; game making; Research through Design; design diaries; teaching ethics*

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Introduction

Design can be seen as a quintessentially ethical process that significantly impacts our collective futures (Devon & Poel, 2004), as designers determine ethical costs and benefits of their actions: who the beneficiaries of design will be, whose agenda and interests define design problems, and which design artifacts will be produced.

Design practice is an ethical process, and the design of products, services, and experiences that are powered or augmented by digital technology (hereafter "technology design"), is a future-shaping, fast-moving field with a tremendous influence on people's lives that presents an even clearer case for the need of some formalized "design ethics". First introduced by Papanek (1971) and later reflected on by Chan (2018), the field of "design ethics" explicitly addresses the ethical responsibility that exists within the practice of design due to potential damages to the environment and influence on people's lives. Unfortunately, Papanek's designerly concerns with ethics are not yet an established component of design education (Ramirez, 2006). Reflections on the 'central role of the economic factor as the almost exclusive evaluation criterion' which 'leads one to consider the user as a mere customer' (Findeli, 2001, p. 6). The consequences of such a view for design ethics leave much unaddressed. Design ethics are often not integrated (Sonneveld, 2016) or reduced to an interest in sustainability and the environmental aspects of production rather than directed at addressing the ethical implications of core design practices (Haug, 2016).

Nowadays, the ethical consequences of technology design are still hardly ever considered before new products and services are marketed for mass adoption (Monteiro, 2020; Shedroff, 2009). The recent meteoric rise of large language models such as ChatGPT or image generators such as MidJourney, present problematic profiles (Appel et al., 2023; Bender et al., 2021; Wilkins, 2023), and yet all are rapidly embedded into the fabric of everyday life. These are but the latest examples of techno-centric practices where economic advantages outweigh ethical concerns.

Despite broad societal impact, design ethics have not gained enough traction in academia (Chan, 2018) or design education (Findeli, 2001; Sonneveld, 2016; Haug, 2016), and research into the ethical consequences of technology design remains scarce in comparison to research in designing new technology. Design practitioners have called for approaches that can provide ethical guidance (Zollman, 2021) for 'reflexive design practices in research, education as well as in the public and private sectors' (Ozkaramanli et al., 2022, p1), to complement normative, check-list-type approaches that are borrowed from disciplines with epistemological statutes different from those of design (Cross, 2001). As a response to the call for framing design ethics in education, we have developed a pedagogical approach to enable students to learn about design ethics that rests on knowledge production, problem exploration and documented reflections. We have applied this approach to teaching design ethics in a master's course: the following sections describe the rationale for the course design, the course activities, and preliminary insights and reflections from running the course.

Course Design Rationale

The graduate course on "Ethics in Digital Service Innovation" (EDSI) is a 7.5 ECTS credits part-time, on-campus course, that runs over for eight weeks. It is a curricular course for first year students in Digital Service Innovation (DSI) at Halmstad University in Sweden. The EDSI course follows program courses on academic communication, services in the digital society, emergent themes, and research methods in informatics. No assumption is made in the course that students would have prior knowledge of design and design theory other than what they learned in the aforementioned courses.

This master's program prepares students to work on the design and development of smart and sustainable digital services by bringing a researcher's mindset into such challenges. Ethics are an important subject in the DSI program: it is a core theme driving activities and reflections and a learning goal that is addressed in all program courses. As such, EDSI was developed with an explicit focus on teaching students how to work with complex ethical issues within the context of designing services. Since research thinking is a key characteristic of this education, and artificial intelligence technologies have been dominating recent technology narratives, these are key points that were addressed in lectures, seminars, and assignments.

EDSI promotes learning through design experiments rather than through the delivery of a finished design concept. The course's pedagogical structure rested on three specific methodologies brought together to provide the students with a well-rounded framework:

- A Research through Design (RtD) theoretical framing where prototyping is used to explore a specific problem space (Giaccardi, 2019);
- An experiential, game-based learning method, called the Design Games Framework (DGF), to support exploration through tabletop game making (Hylving et al., 2023; Resmini, 2022);
- Design diaries as annotated portfolios, representing knowledge through ultimate particulars (Gaver, 2012), and as a best practice for RtD inquiries (Bardzell et al., 2016).

Research through Design for Exploring Design Ethics

RtD is using design as the method of inquiry in a research process meant to generate new knowledge (Zimmerman & Forlizzi, 2014), which is the reason why RtD is a key component of the EDSI. RtD differs from the typical, reductive view of design as a problem-solving activity. Previous work describes design as a polymorphous activity (Lawson, 2006) that addresses problem spaces ranging from the well-defined (Simon, 1969) to the ill-defined and wicked (Buchanan, 1992; Rittel & Webber, 1973). In these multiform design activities, research may be one part of the solution-producing process, as it is, for example, formalized in goal-directed (Cooper et al., 2014) or activity-directed processes (*Design Council*, 2007), or the end goal, as it is in RtD.

In design education, the use of RtD shifts the focus from the production of a solution to the production of new knowledge gained through attempting to solve the problem. This shift is a fundamental component of the learning process if students are to be able to practice reflection-in-action which is emblematic of any skilled designer (Schon, 1968, p3). As design ethics are particularly challenging to address, RtD facilitates investigating ethical issues by replacing the weight of responsibility that is associated with producing a solution with a knowledge-generating activity where designerly exploration through ideation and prototyping is the primary goal, thus reinforcing the reflective nature of the process.

Prototyping design concepts using an RtD methodology can be used for exploring areas of concern (Giaccardi, 2019), enabling designers to explore the design space and iteratively challenge and reframe the design problem. Each subsequent research artifact contributes new knowledge (Zimmerman & Forlizzi, 2014) and embodies a particular aspect of the dialectic relationship between problem and solution (Beckett, 2017; Lindenfalk & Resmini, 2022). Further, artifacts that are prototyped in RtD processes have been found to have a stabilizing effect, helping achieve consensus on the issue at hand among designers and non-designers (Giaccardi, 2019; Sanders & Stappers, 2014).

Several methodologies can be adopted to pursue RtD activities through designerly exploration of a problem space in lab, field, and showroom settings (Koskinen et al., 2011; Zimmerman & Forlizzi, 2014). As a match for the complexity and sensitive nature of design ethics problems, we chose game making as an appropriate, lab-type prototyping activity.

Game Making as Exploration

Ethics is intrinsically difficult to teach due to the vastness, ambiguity, and unpredictability of ethical consequences (Johnson & Others, 2010; Zandvoort et al., 2000). The international and diverse student body that characterizes DSI and the EDSI course introduced additional layers of complexity due to varied cultural and educational backgrounds that magnified differences in how students thought about ethical issues. A game-based and hands-on approach was deemed to provide an effective way to lower the barrier to entry and create a learning environment that could engage as many students as possible in ethical considerations.

Games are a well-established learning tool (McGonigal, 2011; Squire & Jenkins, 2003; Tobias et al., 2014). Experience-based learning approaches use games because their participative and interactive nature can “help increase immersion” (Trybus, 2015), “appeal to different learning styles”, and “encourage collaborative problem solving” (Boyle, 2011, p. 3). Specific game-based learning methods have been created that effectively redesign learning tasks “from the ground up” to “tak[e] advantage of the unique affordances of games” and make sure that the result is “a good learning task but also a good game”(Plass et al., 2020, p. 5). This balance is necessary to ensure that a real-life problem can become an engaging learning experience that moves players towards their goals while allowing them to “choos[e] [their] actions and experienc[e] the consequences of those actions along the way” (Trybus, 2015).

Engaging with game-based learning has two benefits: (1) progression through the game entails learning what the game does, why it does it, and to achieve which goals; (2) engaging with the game is a risk-free activity in which making mistakes carries no real consequences (Crawford, 1984; Trybus, 2015). Players experiment and explore the consequences of specific choices and behaviors without fearing physical, psychological, social, or economic harm.

In consideration of the RtD approach and the use of prototyping as an exploratory tool, EDSI adopted the Design Games Framework (Hylving et al., 2023; Resmini, 2022) (DGF) as the methodology to be used to engage in ethical problem exploration by making tabletop games. The DGF emphasizes learning about a problem space through designing and making a game rather than through playing and reflecting on an existing game. It also favors tabletop games and the embodied, situated multimodal exploration (Massaro, 2012) and material anchoring (Hutchins, 2005) that physical, low fidelity prototyping allows.

Multimodal exploration places a strong emphasis on people's active and situated choices in aligning various modes into an ensemble and exploiting different modes to make meaning (Jewitt, 2013). Similarly, a multimodal artifact such as a game prototype, can be seen as the material outcome or trace of what ethical point is seen as important to attend to and to make salient in respect to the task of game prototyping. A multimodal exploration therefore allows a diverse group of students to actively select, negotiate, and align their rationale for communicating a certain ethical point through specific game features and dynamics.

The DGF offers a manageable way to deconstruct games by means of three sets of elements that work as a system and address the logical and procedural constraints the game places on play (formal set), the game's own narrative (dramatic set), and the environment players will inhabit (spatial set) (Resmini, 2022). These elements are applied in three consecutive phases which mark a reflection-to-generation process in line with the experiential learning progression (Moon, 2013): *Play*, in which selected games are played, deconstructed in their formal, dramatic, and spatial elements, and the play experience discussed; *Remix*, in which existing games are modified by recasting one or more selected formal, dramatic, or spatial elements; *Design*, in which entirely new games are created that introduce novel formal, dramatic, or spatial elements in response to the problem space at hand. This emphasis on the generative aspect of designing and making a game is an important distinctive element of the DGF that merges experiential learning with game-based learning approaches (Boller & Kapp, 2017; Kapp et al., 2014).

The DGF has been used as a prototyping approach to foster exploration and engagement in executive education, industry workshops, and graduate courses (Hylving et al., 2023; Resmini, 2022), and the EDSI students were introduced to it in a series of complementary extra-curricular DSI activities, called "Core Theme" (CT), carried out in the months prior to the course.

Design Diaries

Documentation is a cornerstone of RtD since documentation captures the prototyping, motivation and reflections that constitute the knowledge that RtD generates (Bardzell et al., 2016; Zimmerman & Forlizzi, 2014; Zimmerman et al., 2010). RtD documentation has been referred to as annotated portfolios (Bowers, 2012), speech acts (Bardzell et al., 2016) and workbooks (Gaver, 2011), that typically capture and represent RtD-created knowledge in a chronological sequence. To elevate the importance of time-stamping RtD activities as well as the reflective character of RtD documentation, and inspired by diary study methods, we used the term "design diary" to refer to RtD documentation in the context of the EDSI course. Design diaries were used for the secondary purpose of collecting the data that is the basis of this paper (Figure 1).

Course Activities

Twenty-seven students attended the EDSI course described in this paper. These students came from a variety of countries (twenty-one of them from outside the European Union and six from EU countries), and from different backgrounds. A majority of students had undergraduate and graduate degrees in engineering; some a degree in marketing, business, or finance; and only three had degrees in design. Figure 1 illustrates the architecture of the EDSI course and the individual activities in relation to each other and to the chronological weekly progression.

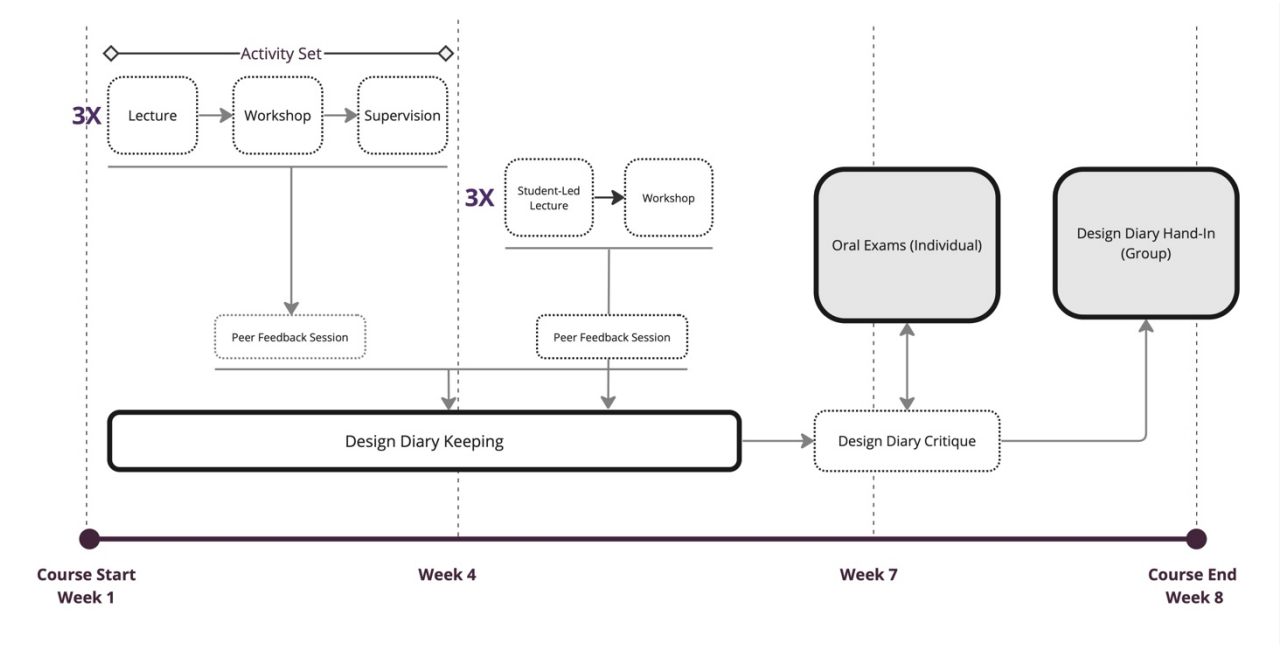


Figure 1. Diagram illustrating the course components, their relationships, and the course timeline

The students worked in self-assigned groups of up to four students all throughout the course. Group work was carried out in three weekly iterations tied to the course *activity sets* and additionally three weekly pairings of student-led lectures and workshops (Figure 1). The diary keeping, diary hand-in, and oral exam activities are outlined in black to indicate that these are also data collection activities that produced the data for this paper.

Instructor-led activity sets had the explicit purpose of tying ethical issues to the practical, experiential activities of RtD through DGF game making. Each activity set consisted of a two-hour, in-person lecture; a workshop session; and an online supervision session. All lectures introduced theory dealing with ethics and design ethics, and ended with a slide presenting a specific ethical problem that students were to work with during the rest of that week. The lecture was followed by an in-person workshop session where groups worked on Remixing games to explore that activity set's ethical problem as presented in the lecture. An online supervision session followed the workshop, typically planned for the day after the workshop. During supervision, the students had the opportunity to show their progress, their exploratory game-making efforts, and receive further feedback from the course instructors.

The activity sets function as design sprints that tackled a new ethical issue weekly, and encourage students not to get attached to any particular prototype. This weekly pacing built into the course structure was also meant to help clarify that they were not asked to keep iterating a concept until they had a solution, but rather work on their games in an exploratory, low fidelity fashion, focusing on unpacking the problems underlying the ethical issue presented by the current activity set.

To further reinforce this crucial exploratory aspect, it was decided to limit the game-making activities to the Remix phase of the DGF. The students could thus focus on modifying an existing game rather than create a new one; manage a smaller problem space; more easily begin prototyping; and could better focus on both the knowledge creation aspects and the balancing of ethical problems and game elements. Concerns about the unfamiliarity of most of the class with design methods and activities and known issues in knowledge transfer between the Remix and Design phases of the DGF in higher education settings (Hylving et al., 2023) were also factored in.

Groups were instructed to keep one single collaborative design diary to capture their RtD work as they progressed through the course. A few requirements were set to guide and standardize the diary-keeping activities: the diary had to be in Google Doc format and shared with the course instructors (commenting privileges only); the diary had to be continuously updated; the course instructors would check on the diary now and then throughout the course. In describing their work in connection to the individual activity sets, they were also specifically asked to detail:

1. The initial goal of each design and how this related to the theory presented in the given activity set and the course as a whole;

2. A short visual presentation of key features of the design and a short explanation and motivation of how these features related to the goal of the week;
3. A short section, including visual material, where they presented features of the design that were changed (for instance in peer-review sessions), along with a brief motivation for the changes.

Lectures, Workshops and Supervisions

The first lecture (L1) introduced the epistemological and philosophical frailties constituting the basis of the development of AI-driven technologies and relied primarily on Crawford's "Atlas of AI" (Crawford, 2021), and Stahl's "An Ecosystem Perspective on the Ethics of AI" (2021). The lecture presented the historical and cross-sectional account of technology formation to illustrate how different geographical locations and groups of people have been affected by the development and use of AI across time and space. The lecture particularly focused on foregrounding the often backgrounded environmental and human-labor narratives to emphasize how different locations, people, and histories are interconnected in complex ways. The ethical dilemma presented at the end of this lecture was: "The making of AI is environmentally damaging, but once we have it in place, we can contribute to environmental sustainability causes in unprecedented ways".

The second lecture (L2) introduced concepts relating to ethics and the design of information technology. These concepts included: scripting, where technology has designed within it the ways which it is meant to be used (Verbeek, 2006); in-action ethics that address the gap between ethical guidelines and in-situ ethical challenges that arise from user involvement (Frauenberger et al., 2016); and ethical issues connected to use of big data (Goodman, 2014). The ethical dilemma presented at the end of this lecture was: "Fairness: AI systems should not discriminate against groups, communities or individuals".

The third lecture (L3) introduced issues related to research ethics and considerations when working with human participants. The students were given a brief overview of the ethical review processes for research studies involving human subjects as well as current EU and Swedish legislation that govern research activities. During the second hour of lecture, the class participated in a guided discussion reflecting on gender, human dignity, and how assumptions made during the study design process can affect the generalizability of research outcomes. For the workshop assignment, the students were asked to consider the scenario: "A researcher is applying for an ethics permit wherein an AI tool would mine a hospital's local patient records to predict which patients are most likely to become deceased within six months".

Each lecture was followed by an in-person hands-on workshop session. The groups were tasked with Remixing one of two games chosen by the instructors, "Fluxx" or "Werewolf" (described in the Prototyping section below) to address the specific ethical challenge discussed as part of that activity set and presented at the end of that week's lecture. The instructors mentored the groups as they designed their tabletop game prototypes, giving tailored feedback and providing both practical and theoretical guidance as needed. Since all groups worked in the same physical space, the students could listen in on feedback provided to other groups, potentially presenting them with additional knowledge, methods, or techniques they could use for their own exploration, and the instructors could quickly identify common issues and immediately address the whole class if necessary.

An online supervision session followed the workshop, typically on the following day. During supervision, the groups had the opportunity to further show and discuss their progress on game making with the course instructors, ask questions, and receive additional feedback.

Student-led Sessions and Peer Feedback Sessions

Teacher-led activity sets were complemented by student-led sessions (Figure 1). In these sessions, groups were asked to select one ethical issue they considered relevant to the course, and to deliver a fifteen-minute lecture on that issue. Selected topics included ethics of AI-driven autonomous weapons systems (Gómez de Ágreda, 2020), ethics and diversity in AI policy (Roche et al., 2022), using ChatGPT in education (Tlili et al., 2023), the ethical implications of institutional racism in healthcare (Elias & Paradies, 2021) and assessing gender bias in machine translation (Prates et al., 2020). The fifteen-minute lectures were followed by a class discussion and then by a two-hour workshop where the lecturing group mentored the other groups as they explored the issue further through game-making.

To complement workshops and supervisions, and to encourage collaboration and group cross-pollination, the course structure included two scheduled peer-feedback sessions (Figure 1). In each session, pre-assigned group pairs would playtest each other's game prototypes and trade feedback. The class was encouraged to self-organize additional

sessions as needed, while outcomes of all peer feedback sessions had to be documented in the design diaries (Figure 1).

Prototyping

Prototyping was limited to selecting and remixing one of two predefined games. The two games that were chosen as the initial template the students could choose to work from for their assignments were “Werewolf” and “Fluxx”.

“Werewolf” is a social deduction game (Robertson, 2010) in which players are assigned a secret identity in one of two opposing factions: a minority group, the werewolves, who know each other, and a majority group, the villagers, who possess no information on other players. In alternate rounds of night and day, the factions try to eliminate players from the opposing group: villagers win when all werewolves have been ousted; werewolves win when there is numeric parity between the two factions. “Werewolf” is a game based on information asymmetry and it is particularly suited to expose bias and prejudice since players possess little to no information to base their decisions on.

“Fluxx” is a 1997 card game by American company Looney Labs in which both the basic rules, draw one card then play one card, and the winning conditions are constantly altered by the cards played by the players. The absence of a stable goal and the way game mechanics can be changed, added, or removed by any player by playing a card, makes “Fluxx” a game in which embracing chaos often prevails on strategy; thus, a good example of how complexity and unpredictability emerge from rather simple base structures.

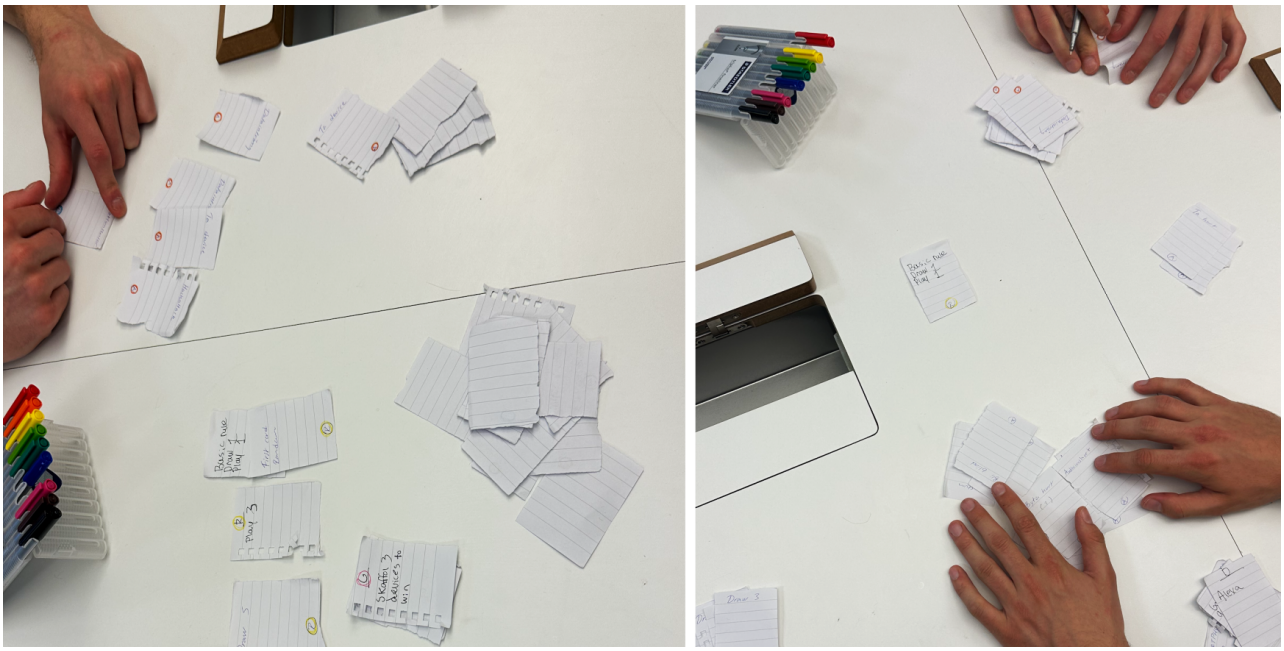


Figure 2. Playtesting a Fluxx remix

These two games were chosen because of the students’ familiarity with them, since they were used repeatedly in the prior CT activities mentioned earlier, and because their abstract and relatively uncomplicated architectures allowed the students to remix them more easily while maintaining that balance between problem space and game that is at the center of DGF explorations and necessary in game-based learning (Plass et al., 2020).

Table 1 contains a synthetic view of the most significant dramatic, formal, and spatial elements of “Werewolf” and “Fluxx”. It should be noted that “outcome” is itself one of the formal elements but has been given more relevance here as the element identifying the “goal” of the game, and thus an important point of reference for the students working on their remixes. Spatial elements are minimal in both games as they are card games with a basic requirement to share a common space (enclosure), to hide one’s cards or identities (separation), and that use sequence to regulate gameplay.

Table 1. A synthetic view of the main dramatic, formal, and spatial elements of Werewolf and Fluxx

	Outcome	Dramatic Elements	Formal Elements	Spatial Elements
Werewolf	The villagers win if they eliminate all werewolves from the game. The werewolves win if their number matches that of villagers	Premise: werewolves hunt people to feed Settings: a (medieval) village is hunting ground for werewolves Story: at night, werewolves hunt. During the day, villagers try to get rid of them Characters: villagers, werewolves, seer (additional characters exist)	Players: minimum 7, engaging in team competition Procedures: day and night cycle, voting processes managed by a game master Rules: every night werewolves claim a victim who is eliminated from play. During the day villagers vote to eliminate someone they believe is a werewolf. Each villager casts one vote Resources: character information, information acquired by werewolves, information acquired by seer, individual vote	Enclosure: circle of players. Players who are eliminated are requested to take a step back Nesting: werewolves form a secret group within the villagers group Sequence: night/day cycle as two different spaces for action
Fluxx	Achieve whatever "goal" is the current "goal"	Premise: - Settings: the world today (original Fluxx) Story: - Characters: -	Players: 2-6, engaging in multilateral competition Procedures: draw one card, play one card Rules: changing based on cards played Resources: cards representing goals, rules, actions, keepers Challenges: rules change constantly, goals change constantly, chance	Separation: player's hand is kept secret Sequence: turn-based play

Table 2 and Table 3 provide selected samples of two "Werewolf" and two "Fluxx" Remixes, highlighting the changes the students introduced to specific elements along with motivation they provided in their design diaries. The labels L1, L2, and L3 refer to the respective activity set lectures.

Table 2. Examples of changed outcomes, dramatic, formal, and spatial elements in "Werewolf" Remixes, with motivation

	Outcome	**Dramatic Elements**	**Formal Elements**	**Spatial Elements**
Remix Group 1 (L1)	Players are either AI or human and can switch sides. When a side reaches a certain number of players, they win	Settings: The future Premise: AI vs humans Characters: AIs and humans	Rules: Replace eliminating players to players switching between the AI and human side	Same as original game
Motivation	Highlight the loss of jobs due to wider adoption of AI	Paint a futuristic world that focuses on the human roles and labor powering AI work	Show AI taking over human jobs	-
Remix Group 2 (L3)	Stimulate discussion on the use of sensitive personal data to develop AI in healthcare. The game ends when players feel they have sufficiently explored the issue	Settings: The world today Premise: Researchers apply for ethics permit to make AI tools Story: A committee needs to decide whether or not they will issue an ethics permit	Resources: Cards representing AI-related roles, processes, and artifacts	Same as original game
Motivation	Increase ethical awareness when developing AI tools using patient records	Stimulate discussion and improve the understanding of the ethical implications of developing AI for healthcare	-	Present arguments and open discussion

Table 3. Examples of changed outcomes, dramatic, formal, and spatial elements in "Fluxx" Remixes, with motivation

	Outcome	**Dramatic Elements**	**Formal Elements**	**Spatial Elements**
Remix Group 3 (L3)	Have all stakeholders on your side and obtain ethical permit before everyone else	Settings: The world today Premise: Researchers apply for ethics permit	Resources: cards representing patients, family, hospital, permit committees	Same as original game
Motivation	Illustrate the need for permits, the complexity of getting one, the researcher's lack of control	-	-	-
Remix Group 4 (L2)	Same as the original game	Settings: Transportation	Resources: cards representing means of transport, sustainability goals, rules that award green mobility, actions for carpooling or bike share	Same as original game
Motivation	Illustrate how implementing AI in mobility addresses sustainability challenges	Focus on sustainable transportation to address environmental sustainability with AI	-	-

Examination

Examination had two parts, a group hand-in, and an individual oral exam. The group hand-in consisted of the complete design diaries, containing all the group's prototypes, conversations, reflections, and motivations. The diary

deliverables were uploaded in digital format to the course's online platform and were graded for clarity, strength of argumentation, and for how well they matched ethical issues and design exploration (Plass et al., 2020).

The in-person, individual exam—a legal requirement for higher education course grading in Sweden—consisted of an interview in which each student was asked to critique a diary entry taken another group's design diary drafts. These draft diaries, containing documentation of the work resulting from the teacher-led sessions, were handed in two weeks before the oral exam and only used for the purpose of providing the students with entries to critique. A week before the individual exams, the course instructors made all selected diary entries available to all students, along with several topics of critique. These topics centered on the match between diary content and the ethical issue, and on clarity of argumentation, with specific attention to be paid to how the RtD game-making work was used to explore the ethical problem. During the oral exam, each student was randomly presented with a diary entry to critique and given fifteen minutes to expose their reflections and answer follow-up questions from the instructors.

Outcomes and Reflections

What follows describes the effectiveness of the game-making approach in enabling students to learn about ethics in the design context. The effectiveness of our teaching approach is explained in terms of three overarching outcomes that helped students to (1) move away from normative, check-list approaches to ethical discussions in a design context, (2) Internalize a focus on developing knowledge and awareness instead of solutions, and (3) Replace the metaphoric and surface discussions of ethical issues and explore the deeper, more structural relations of developing sustainable and ethical socio-technical systems. These overarching outcomes also illustrate the effectiveness of game making as a worthwhile approach for teaching ethics on a more conceptual level. They do so by showing game making to be helpful in avoiding "cultural cloning" in teaching moments that are usually a result of normativity and imagined homogeneities (Essad, 2021). Additionally, game making as structured through the DGF is overall in accordance with the calls for recent theory building in adult learning where the emphasis is on engaging the body, emotions and spirit—holistic learning—to construct learning or knowledge (Merriam 2018).

Developing Reflexivity Through Game Making

Perhaps one of the most revealing instances where the students relied on reflexive design practices (Ozkaramanli et al., 2022), was the oral examination. Although a major part of the critique of another group's prototype is considered to be a purely analytical endeavor, the students based this analysis on their own group's practical experience of developing a similar prototype. One of the students explicitly articulated this cognitive process at the beginning of their critique: "I always compared what we had done in our group with other people's prototypes when I prepared for the exam, you know". This was an important moment of reflection on one's own experience happening through the critique of someone else's work.

This reflexive practice provided an opportunity for the students to develop a first-hand understanding of the limitations and possibilities of implementing theory in practice, and a move away from normative, superficial approaches to ethical discussions. For instance, upon revealing the exam prototype to one of the students, they mentioned they had hoped they could get a different prototype to critique as the revealed prototype had the least shortcomings which provided little for criticism. However, when the examiners probed the student asking how to improve the critiqued prototype, the student provided a thorough analysis and concluded by commenting "I actually had more things to say about the prototype than I had thought", revealing an instance of how self-reflexivity had helped the students unpack design ethics through game making.

Probing the students to look beyond a normative picture of what they saw at first sight was possible by leaning on their practical experience of design. To provide the critique, it was not enough to have mastery of the game, or recite ethical issues brought up in lectures. Rather, it was the interconnection of theory and practice that mattered. Upon realizing and practicing this type of critique in the oral exam, the students had one extra week to get back to the group work and finalize their own group prototypes based on their new insights.

Design as Exploration vs Design as Solution Making

A key theme and challenge for RtD is the necessary shift from seeing design as a solutioning activity to design as an exploratory research method. Throughout the various course activities students grappled with this challenge, often oscillating between attempting to "solve" ethical challenges through design or attempting to make a "fun game" to pursuing prototypes that would unpack or even exacerbate the ethical issues, in critical design fashion (Flanagan, 2009; Malpass, 2019). A key point in the work of many students was to define the effect that playing the games would

have on the players. In general, students produced better prototypes in an exploratory fashion when they targeted outcomes such as problematization, provocation, and discussion, rather than solving ethical problems.

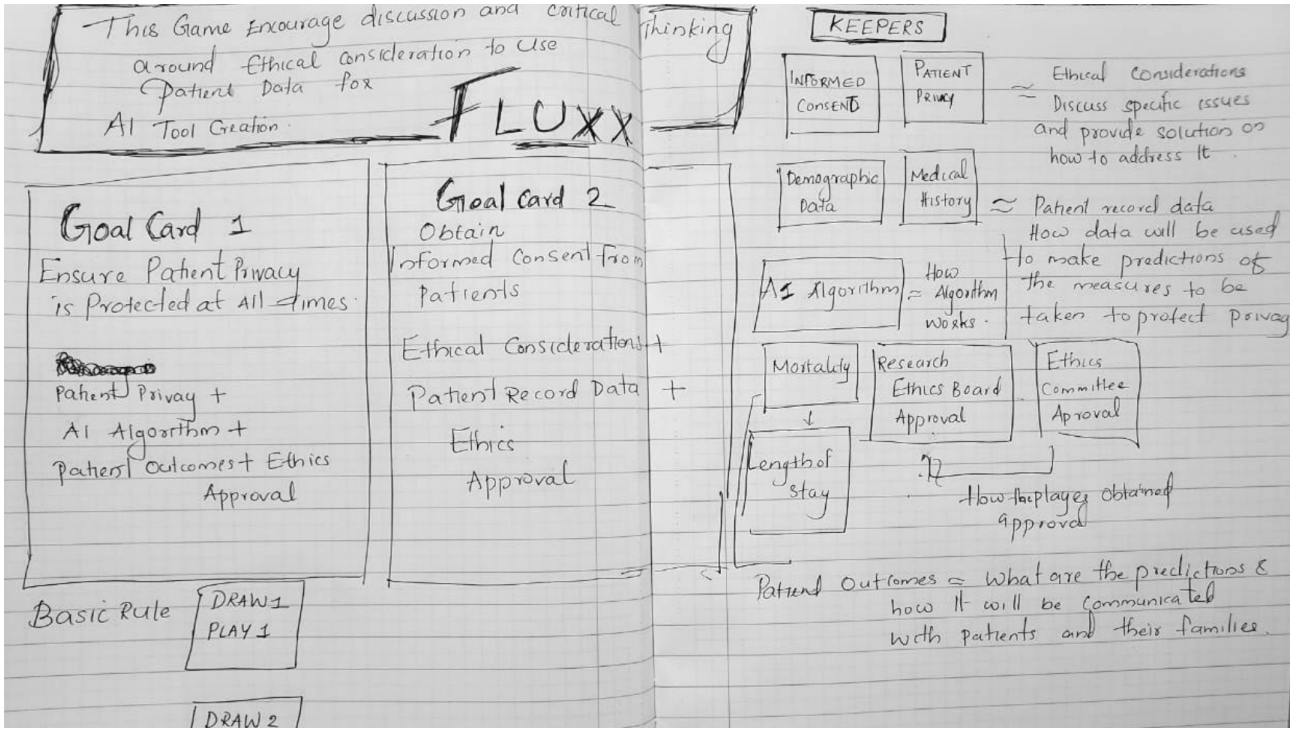


Figure 3. FLUXX remix. Early prototype

For example, one group attempted to remix "Fluxx" to educate players on how AI can enable sustainable transportation choices (Table 3, Group 4). These students configured the outcome so that the win condition would align with using autonomous, public, electrified transportation and "players [would] learn the significant role of AI in optimizing sustainable transportation". However, the ethical challenge they were asked to address was the dissonance between the sustainability costs of making AI and ideas of AI saving humanity by enabling sustainable living: this challenge was not addressed in their RtD game-making sprint, since the group fixed their attention on using the game to promote a solution.

Iterative supervision and tutoring to help shift the students' focus towards knowledge production rather than the concretization of a particular solution is essential in addressing design ethics in the context of exploratory RtD processes. In the example given above, the group was advised to consider and address the loss of human agency that may come from automating away transportation, and the environmental cost attached to running AI algorithms and producing electric vehicles, to move forward and away from the solution space. This led the group to revise their prototyping approach in the follow-up activity sets and considerably improve their knowledge production process by moving from solving to exploring issues of consent and privacy as they worked with the challenge of using data to train medical AI tools (Figure 3).

Moving beyond dramatic elements in DGF

Dramatic elements (settings, premise, characters, story in the DGF) have a visceral impact on players. While "Werewolf" immersed students in medieval fantasy, the "Fluxx" variant used in the course did not rely on any pronounced narrative. Groups working with "Werewolf" would more often get stuck iterating over its dramatic elements, especially during the initial stages, ignoring design opportunities offered by the formal and spatial elements and ending with poorly balanced prototypes where problem and game did not fit each other. When pushed to go beyond dramatic elements, groups found vastly more interesting opportunities in remixing the formal and spatial elements, as shown in the example above (Figure 3), starting a progression loop in which subsequent iterations deepened the understanding of the problem space and consolidated the game aspects of the remix.

The DGF gave students a structured way to go beyond the facade of the games, so that they could leverage the games' characteristics (Table 1) to explore the various problem spaces presented in the weekly activity sets. The use of a methodology the students could refer to allowed them to experientially identify and formalize their own learning

goals beyond those formulated by the instructors and discuss design ethics problems by inscribing them in the context of a design game that could be further used for reflective and generative purposes in the subsequent sprints.

Conclusions

Teaching design ethics is a challenging activity due to the complexities and ambiguities of ethical issues and the perceived lack of responsibility of designers. In this paper we have used RtD, DGF game making, and design diaries to teach master's students in Digital Service Innovation how to engage with ethical issues related to technology design. We found that exploration of the problem space through the remix of tabletop games provided a rich educational setting that facilitated reflexivity and enabled students to shift from a narrow solutionist mindset to a more appropriate and critical mindset. The provision of a consolidated framework for exploration and knowledge production through game making, supported by supervision, enabled students to deconstruct the wicked problems they were given, move beyond cosmetic attempts to tackle core issues, and build a better understanding of the ethical implications of technology design as the course progressed. We recommend design educators to implement game-making activities in the teaching of design ethics to enable students develop more ethical design practices.

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